



Carnegie Mellon University
Master of
Software Engineering

MVC, data model

LG Architecture Training Program

Paulo Merson

Agenda

Module views

View showing module decomposition

View showing module use

Layered style

➔ MVC style

Data model/domain model

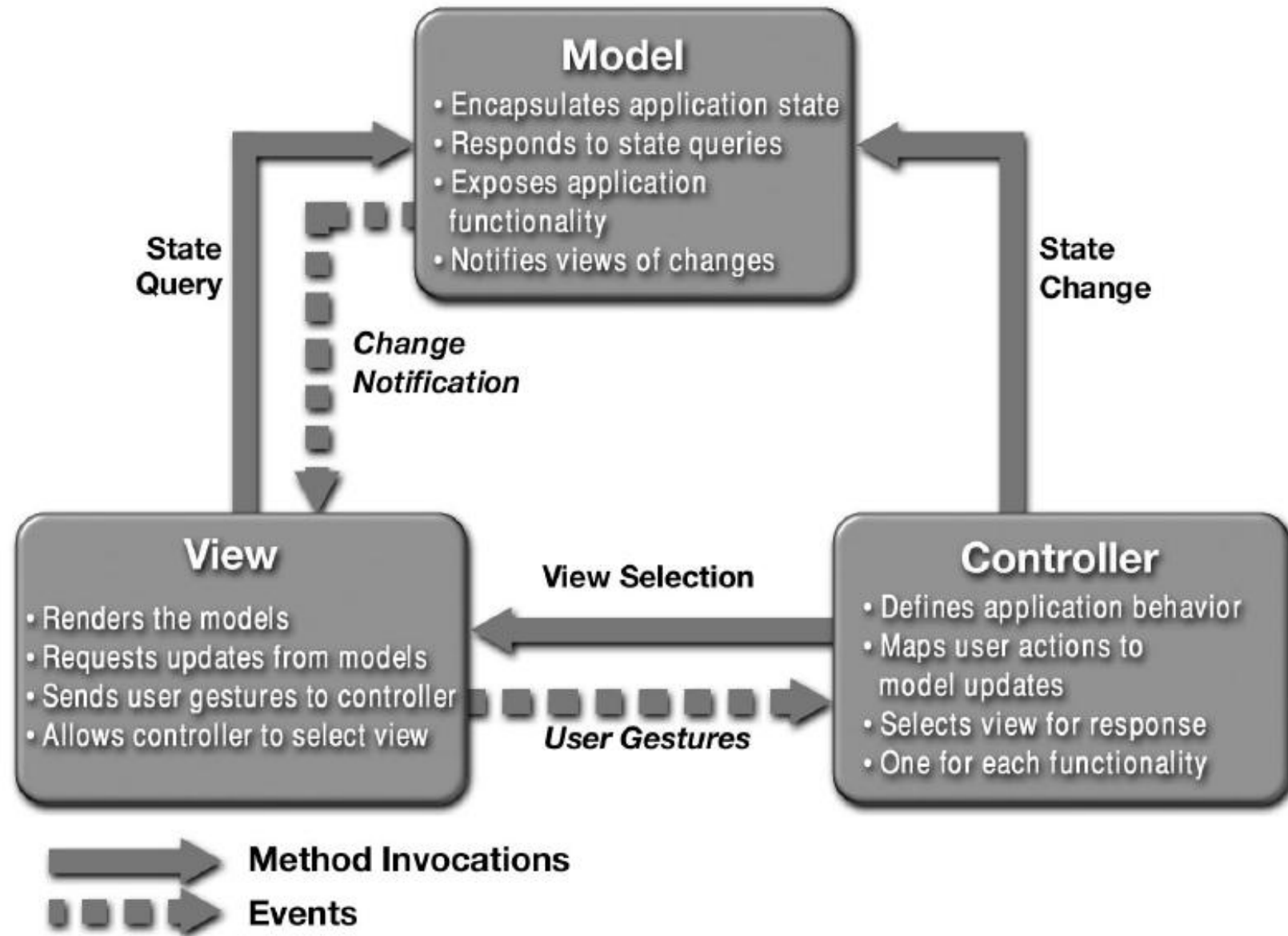
Hexagonal architecture

Microkernel pattern

Model-View-Controller style

- **Elements:**
 - **Model:** encapsulates data (usually persisted data)
 - **View:** UI that exhibits data and captures user actions
 - **Controller:** handles user actions and executes the business logic that may involve changing data
- **Relations:** see next slide
- **(Typical) constraints:**
 - View depends on Model and may depend on Controller
 - Controller depends on Model and may depend on View
 - Model does not depend on View or Controller

How MVC works



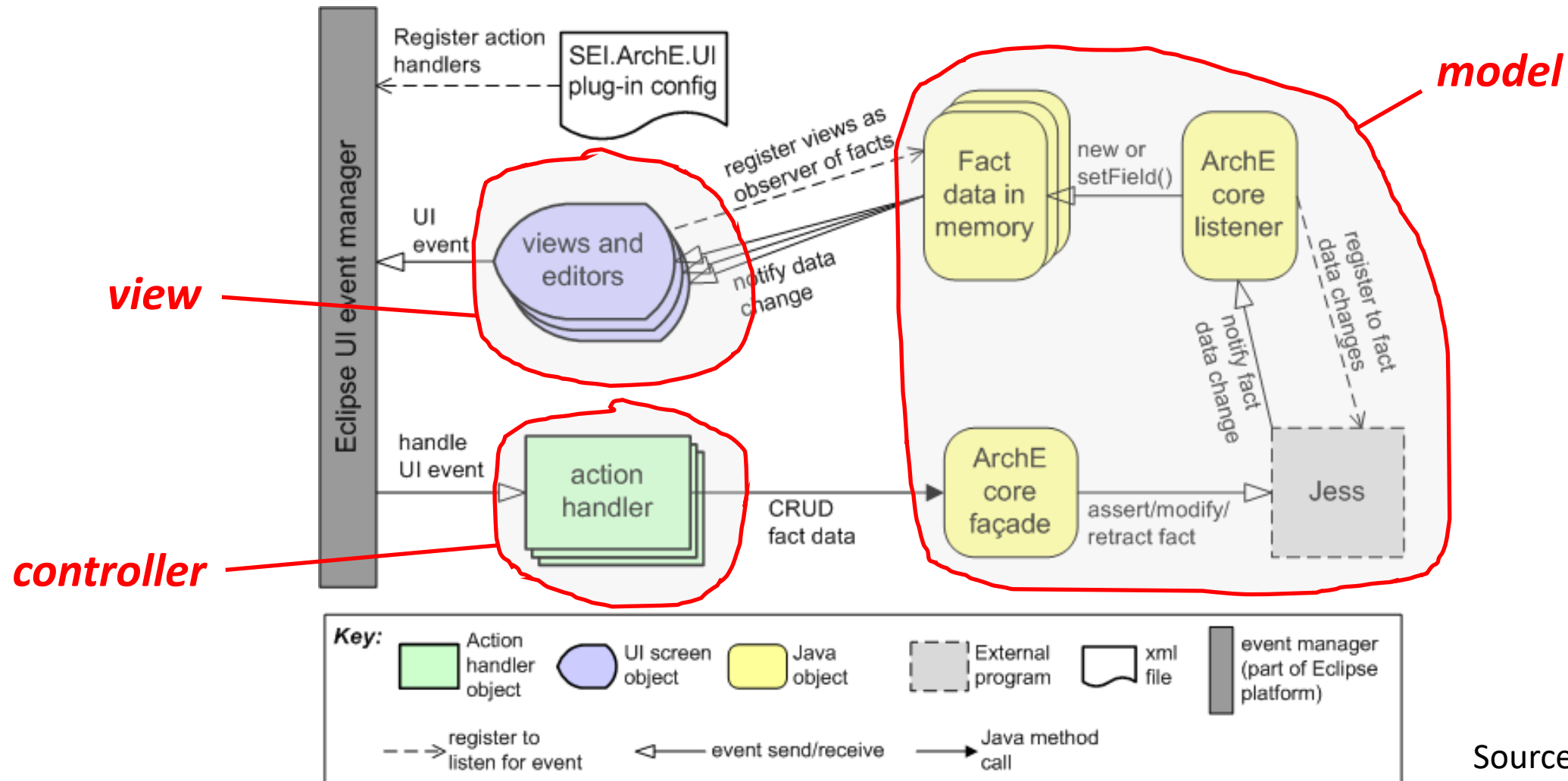
Source: I. Singh et al. *Designing Enterprise Applications with the J2EE Platform, Second Edition*. Addison-Wesley, 2002.

MVC has several variations

- It was originally created for Smalltalk programs
- It was commonly used in GUI applications (i.e., Java Swing)
- It's widely used in Web applications today
- Examples of frameworks
 - Spring MVC
 - ASP.NET MVC Framework
 - Angular
 - Ruby on Rails
 - Django* (Python)

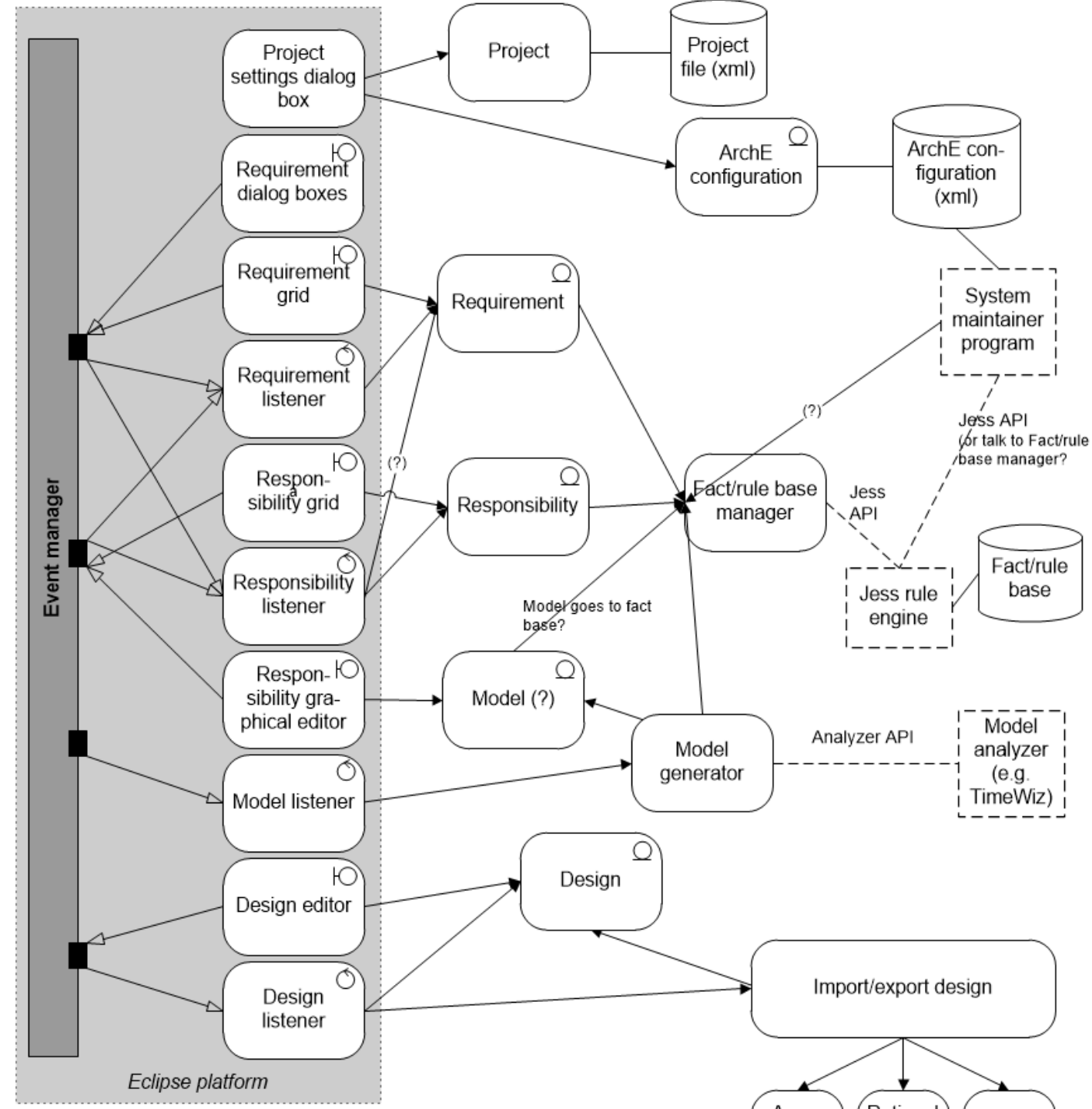
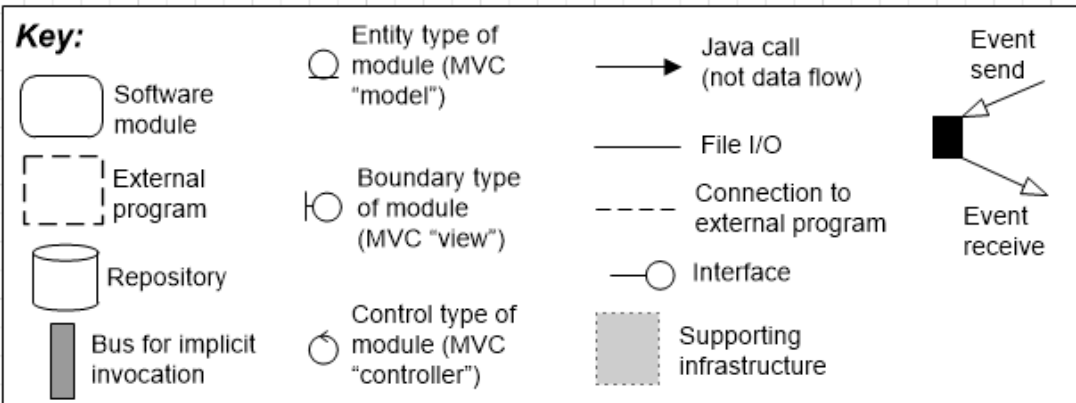
* See <https://stackoverflow.com/a/71664709/317522> for a notable Django MVC quirk

MVC style example



Source: [Clements10]

MVC – same example, an early diagram



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➔ **Data model/domain model**

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Data model

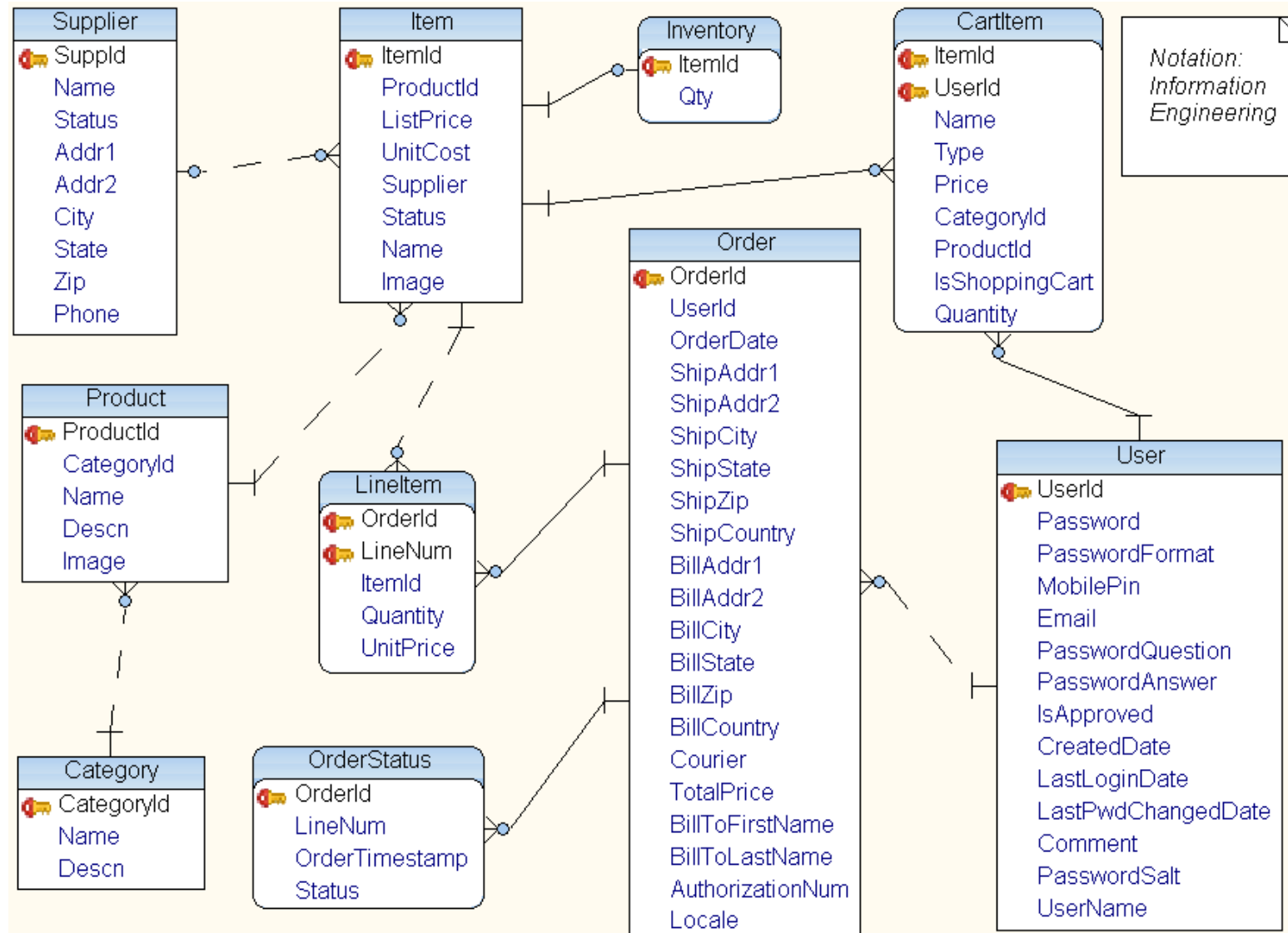
- A (traditional) data model shows the structure of data manipulated in the system
- **Elements**: data entities (persisted domain elements)
- **Relations**:
 - 1:1, 1:n and n:n relationships
 - Generalization/specialization
 - Aggregation

*If data model is an architecture view,
is it a module view? Allocation view?
Something else?*

*The answer has no
practical implications, but
if you're interested, see
Section 9.1 of [Merson09]*

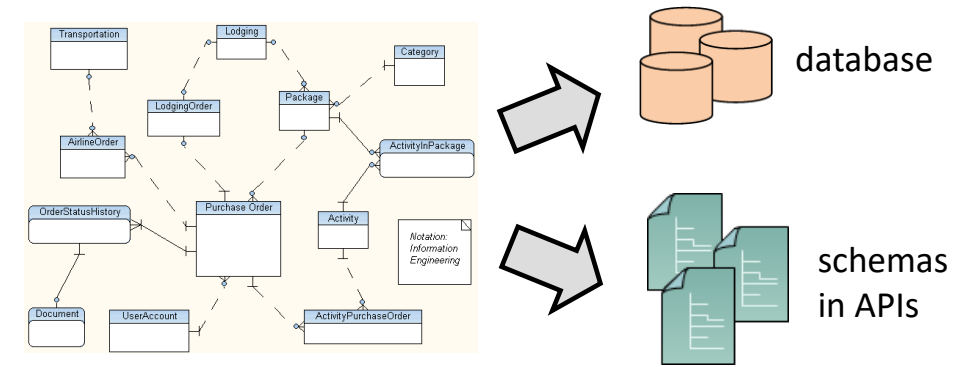


Data Model example – Pet Shop .NET application



What is the data model good for?

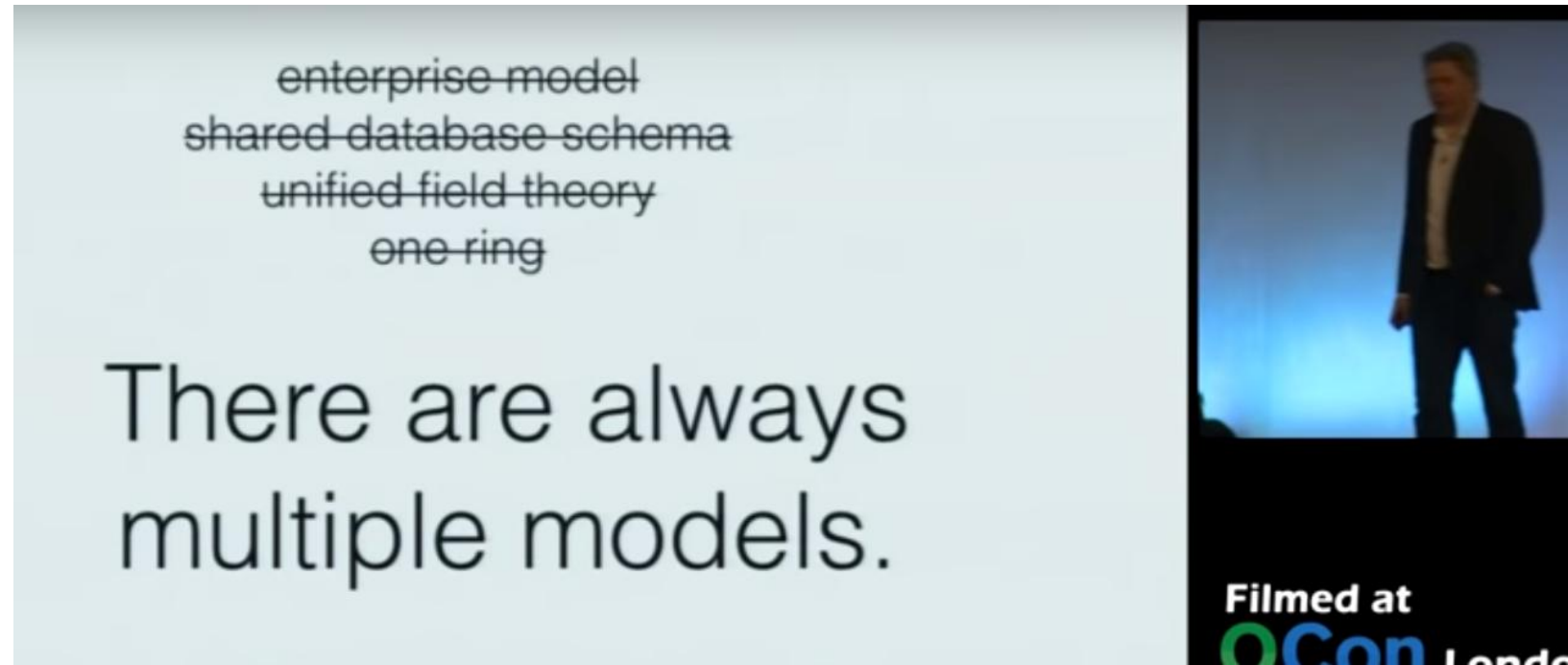
- Blueprint for:
 - Physical database
 - Schemas used in APIs
- Input to normalization
- Assess performance in data-centric systems
 - Merged entities; derived attributes
 - Indexes; lock type and granularity
- Useful reference for developers of data access code
- Modifiability impact analysis



From data model to *domain model*

- The practice has shifted from creating data models to creating **domain models**
- A domain model **focuses on the behavior** (operations) exposed by entities whereas a data model typically only describes the data in each entity
- Another difference is that traditional data models were far more centralized than domain models

We create a domain model for each application, each bounded context

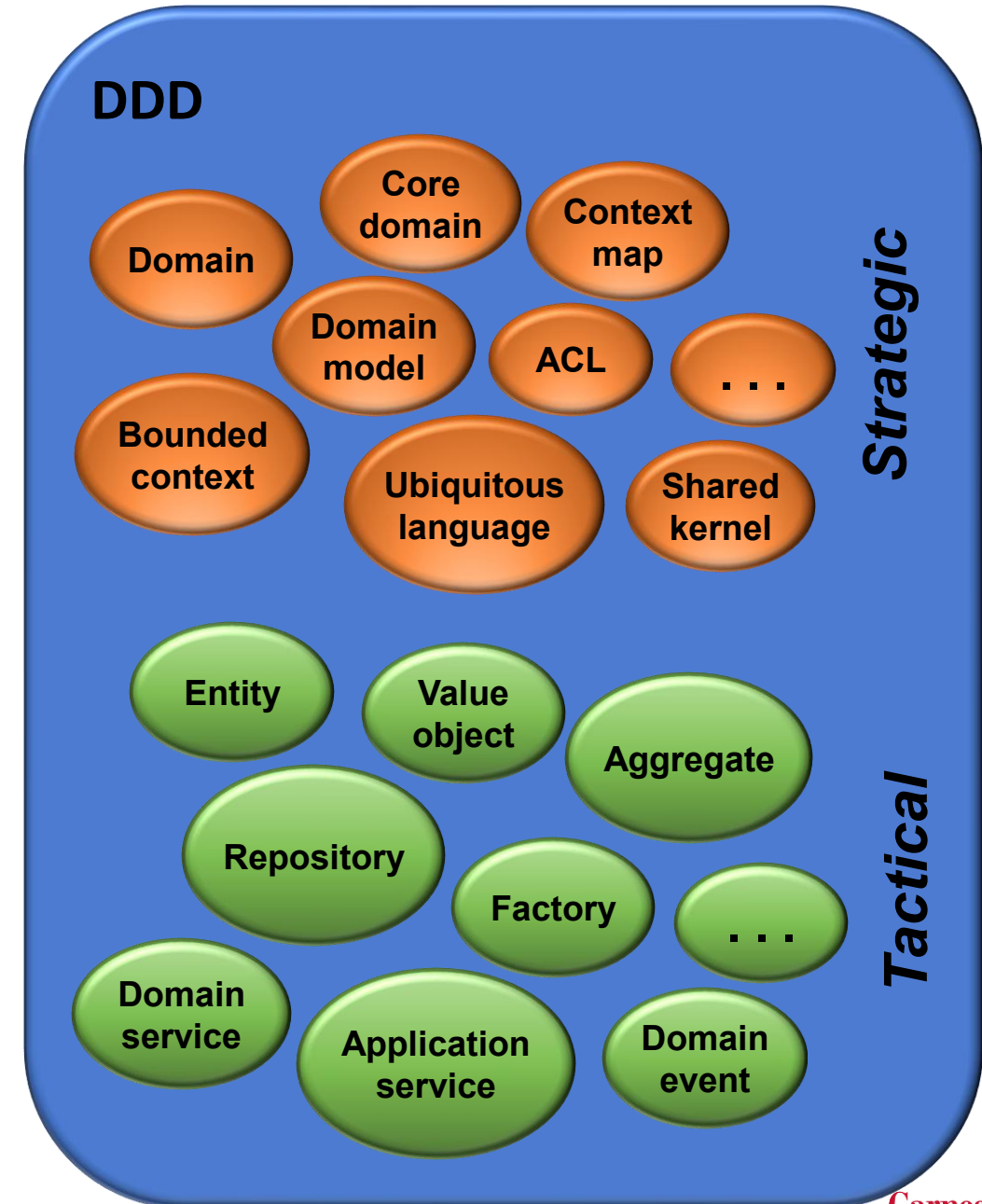


Domain-Driven Design

- DDD is an approach to domain modeling created by Eric Evans
- Beyond entities, it prescribes different types of elements for complex domain applications



Eric Evans



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DDD Data model/domain model

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